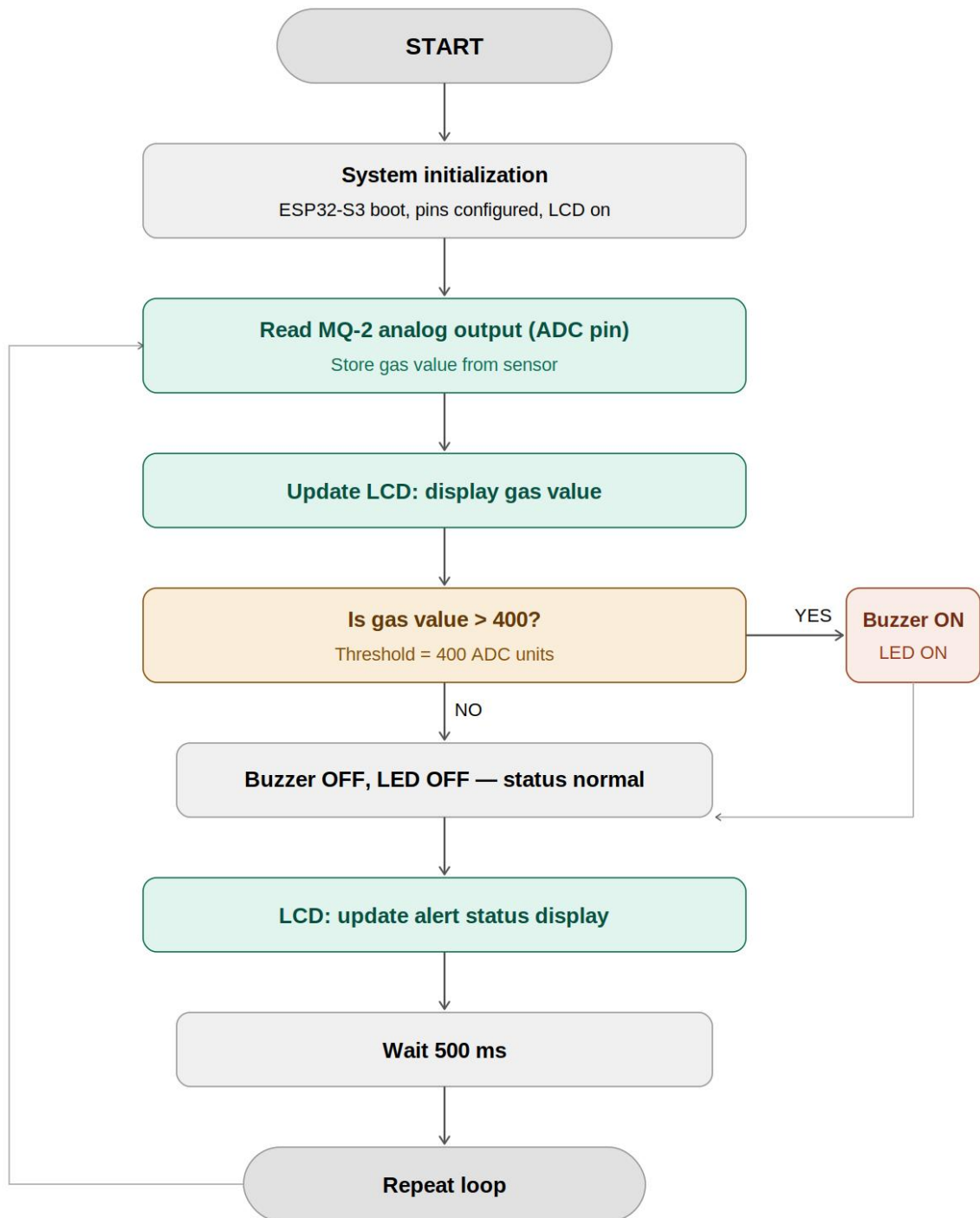


Logic Flowcharts. Smart Home Automation System

The following three flowcharts describe the decision logic implemented in the ESP32-S3 microcontroller firmware for each subsystem. Each chart shows sensor reads, decision points, output actions, timer delays, and the continuous loop structure.

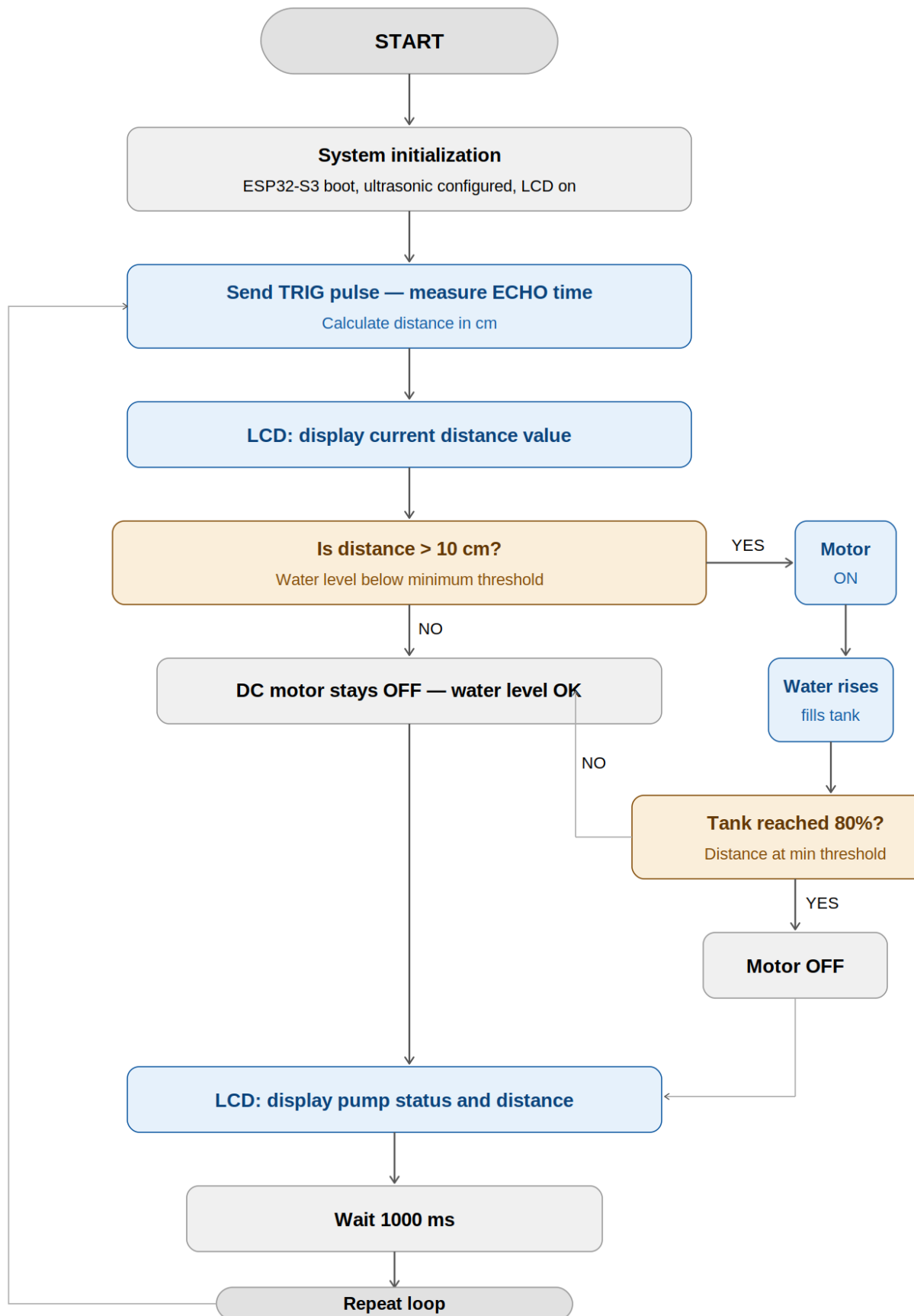
Flowchart 1: Gas Detection and Alert System

Reads MQ-2 sensor every 500 ms. If gas value exceeds threshold of 400 ADC units, buzzer and red LED are activated. LCD displays current gas value and alert status continuously.



Flowchart 2: Water Level Monitoring and Pump Control

Reads ultrasonic sensor every 1000 ms. If distance exceeds 10 cm (water below minimum), DC motor pump turns ON and water begins filling the tank gradually. Once tank reaches 80% capacity (distance reaches minimum threshold), motor turns OFF automatically. LCD shows live distance and pump status.



Flowchart 3: Door Lock Control System

System waits for either a physical push button press or a mobile app command received over Wi-Fi. A button press toggles the door state each time. A mobile OPEN command rotates servo to 90 degrees (unlocked); a CLOSE command returns it to 0 degrees (locked). If no input is detected, system loops back to wait. LCD displays current door state at all times.

